CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 6: Triangles

Step-by-step solutions to the textbook exercises: 29 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 6.1

Q1 Choose the right word: (i) Any two circles are ___. (congruent, similar) (ii) Any two squares are ___. (similar, congruent) (iii) Any two ___ triangles are similar. (isosceles, equilateral) (iv) Two polygons with equal numbers of sides are similar when (a) their matching angles are ___ and (b) their matching sides are ___. (equal, proportional)

SOLUTION

Similar figures have the same shape but not always the same size. Circles and squares can differ in size, so they are similar, not always congruent.

Answer: (i) similar (ii) similar (iii) equilateral (iv) (a) equal (b) proportional

Q2 Give two different examples of a pair of (i) similar figures (ii) non-similar figures.

SOLUTION

(i) Two circles of different radii, and two equilateral triangles of different sides.

(ii) A circle and a square, and a square and a rectangle that is not a square.

Other correct examples are possible.

Q3 Are the quadrilaterals in the textbook figure similar? One is a rhombus $PQRS$ with each side 1.5 cm and no right angle, and the other is a square $ABCD$ with each side 3 cm.

SOLUTION

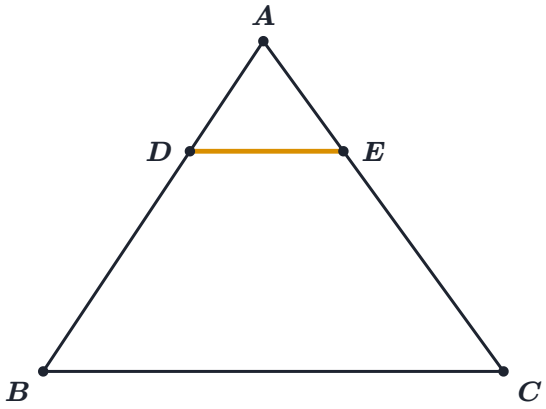
The corresponding sides are proportional: each ratio is $\frac{1.5}{3} = \frac{1}{2}$.

But the corresponding angles are not equal: every angle of the square is 90°, and the angles of the rhombus are not 90°.

Answer: they are not similar

2 Exercise 6.2

Q1 In each case $DE \parallel BC$, with D on AB and E on AC. (i) $AD = 1.5$ cm, $DB = 3$ cm, $AE = 1$ cm. Find EC. (ii) $DB = 7.2$ cm, $AE = 1.8$ cm, $EC = 5.4$ cm. Find AD.



SOLUTION

By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$.

(i) $\frac{1.5}{3} = \frac{1}{EC}$, so $EC = 2$ cm.

(ii) $\frac{AD}{7.2} = \frac{1.8}{5.4}$, so $AD = \frac{7.2}{3} = 2.4$ cm.

Answer: (i) $EC = 2$ cm (ii) $AD = 2.4$ cm

Q2 E and F are points on the sides PQ and PR of $\triangle PQR$. Say whether $EF \parallel QR$ when (i) $PE = 3.9$ cm, $EQ = 3$ cm, $PF = 3.6$ cm, $FR = 2.4$ cm (ii) $PE = 4$ cm, $QE = 4.5$ cm, $PF = 8$ cm, $RF = 9$ cm (iii) $PQ = 1.28$ cm, $PR = 2.56$ cm, $PE = 0.18$ cm, $PF = 0.36$ cm.

SOLUTION

By the converse of the Basic Proportionality Theorem, $EF \parallel QR$ exactly when $\frac{PE}{EQ} = \frac{PF}{FR}$.

(i) $\frac{3.9}{3} = 1.3$ and $\frac{3.6}{2.4} = 1.5$. Not equal, so EF is not parallel to QR.

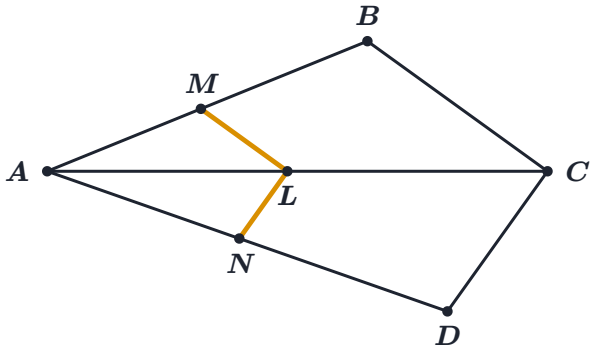
(ii) $\frac{4}{4.5} = \frac{8}{9}$ and $\frac{8}{9}$. Equal, so $EF \parallel QR$.

(iii) $EQ = 1.28 - 0.18 = 1.10$ and $FR = 2.56 - 0.36 = 2.20$. $\frac{0.18}{1.10} = \frac{9}{55}$ and $\frac{0.36}{2.20} = \frac{9}{55}$. Equal, so $EF \parallel QR$.

Answer: (i) no (ii) yes (iii) yes

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Q3 In the figure, L lies on the diagonal AC of quadrilateral $ABCD$, with M on AB and N on AD such that $LM \parallel CB$ and $LN \parallel CD$. Prove that $\frac{AM}{AB} = \frac{AN}{AD}$.



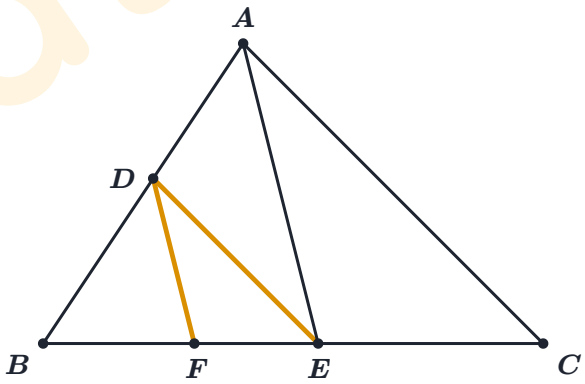
SOLUTION

In $\triangle ABC$, $LM \parallel CB$, so $\frac{AM}{AB} = \frac{AL}{AC}$.

In $\triangle ADC$, $LN \parallel CD$, so $\frac{AN}{AD} = \frac{AL}{AC}$.

Both ratios equal $\frac{AL}{AC}$, so $\frac{AM}{AB} = \frac{AN}{AD}$.

Q4 In the figure, D is on AB and F, E are on BC such that $DE \parallel AC$ and $DF \parallel AE$. Prove that $\frac{BF}{FE} = \frac{BE}{EC}$.



SOLUTION

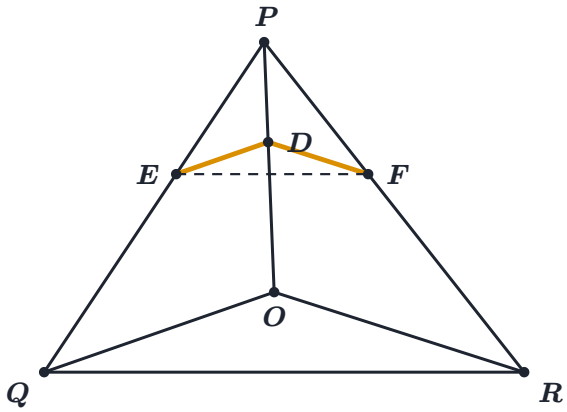
In $\triangle ABC$, $DE \parallel AC$, so $\frac{BD}{DA} = \frac{BE}{EC}$.

In $\triangle ABE$, $DF \parallel AE$, so $\frac{BD}{DA} = \frac{BF}{FE}$.

Hence $\frac{BF}{FE} = \frac{BE}{EC}$.

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Q5 In the figure, O is a point inside $\triangle PQR$, D is on PO, E is on PQ and F is on PR, with $DE \parallel OQ$ and $DF \parallel OR$. Show that $EF \parallel QR$.



SOLUTION

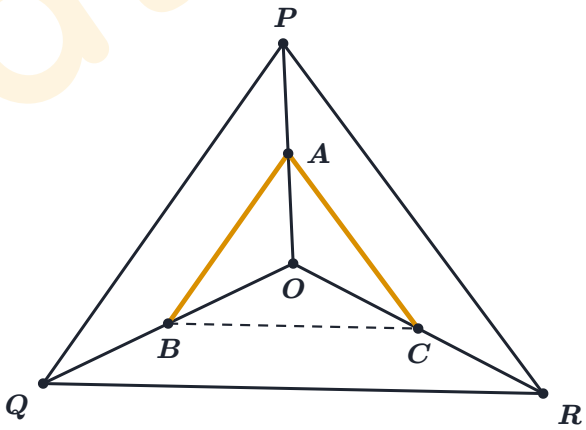
In $\triangle POQ$, $DE \parallel OQ$, so $\frac{PE}{EQ} = \frac{PD}{DO}$.

In $\triangle POR$, $DF \parallel OR$, so $\frac{PF}{FR} = \frac{PD}{DO}$.

So $\frac{PE}{EQ} = \frac{PF}{FR}$ in $\triangle PQR$.

By the converse of the Basic Proportionality Theorem, $EF \parallel QR$.

Q6 In the figure, O is a point inside $\triangle PQR$, and A, B, C are points on OP, OQ, OR such that $AB \parallel PQ$ and $AC \parallel PR$. Show that $BC \parallel QR$.



SOLUTION

In $\triangle OPQ$, $AB \parallel PQ$, so $\frac{OA}{AP} = \frac{OB}{BQ}$.

In $\triangle OPR$, $AC \parallel PR$, so $\frac{OA}{AP} = \frac{OC}{CR}$.

So $\frac{OB}{BQ} = \frac{OC}{CR}$ in $\triangle OQR$.

By the converse of the Basic Proportionality Theorem, $BC \parallel QR$.

Q7 Use the Basic Proportionality Theorem to prove that a line parallel to one side of a triangle and passing through the mid-point of a second side also bisects the third side.

SOLUTION

In $\triangle ABC$, let D be the mid-point of AB and let the line through D parallel to BC meet AC at E.

Since $DE \parallel BC$, $\frac{AD}{DB} = \frac{AE}{EC}$. But $AD = DB$, so $\frac{AE}{EC} = 1$.

Hence $AE = EC$, so E is the mid-point of AC.

Q8 Using the converse of the Basic Proportionality Theorem, prove that the line joining the mid-points of two sides of a triangle is parallel to the third side.

SOLUTION

In $\triangle ABC$, let D and E be the mid-points of AB and AC. Then $\frac{AD}{DB} = 1$ and $\frac{AE}{EC} = 1$.

So $\frac{AD}{DB} = \frac{AE}{EC}$.

By the converse of the Basic Proportionality Theorem, $DE \parallel BC$.

Q9 $ABCD$ is a trapezium with $AB \parallel DC$, and its diagonals meet at O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$.

SOLUTION

Through O draw $EO \parallel AB$, meeting AD at E. Then $EO \parallel DC$ as well.

In $\triangle ADC$, $EO \parallel DC$, so $\frac{AE}{ED} = \frac{AO}{OC}$.

In $\triangle DAB$, $EO \parallel AB$, so $\frac{DE}{EA} = \frac{DO}{OB}$, that is $\frac{AE}{ED} = \frac{BO}{OD}$.

So $\frac{AO}{OC} = \frac{BO}{OD}$.

Rearranging, $\frac{AO}{BO} = \frac{CO}{DO}$.

Q10 The diagonals of a quadrilateral $ABCD$ meet at O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that $ABCD$ is a trapezium.

SOLUTION

The given relation can be written as $\frac{AO}{OC} = \frac{BO}{OD}$.

Through O draw $EO \parallel AB$, meeting AD at E. In $\triangle DAB$, $\frac{DE}{EA} = \frac{DO}{OB}$, so $\frac{AE}{ED} = \frac{BO}{OD} = \frac{AO}{OC}$.

In $\triangle ADC$, $\frac{AE}{ED} = \frac{AO}{OC}$, so by the converse of the Basic Proportionality Theorem $EO \parallel DC$.

$AB \parallel EO$ and $EO \parallel DC$, so $AB \parallel DC$ and $ABCD$ is a trapezium.

3 Exercise 6.3

Q1 For each pair of triangles in the textbook figure, say whether they are similar, name the criterion and write the similarity in symbols. The data in the figure is given in the table.

Pair	First triangle	Second triangle
(i)	$\triangle ABC$: $\angle A = 60°$, $\angle B = 80°$, $\angle C = 40°$	$\triangle PQR$: $\angle P = 60°$, $\angle Q = 80°$, $\angle R = 40°$
(ii)	$\triangle ABC$: $AB = 2$, $BC = 2.5$, $AC = 3$	$\triangle PQR$: $QR = 4$, $PR = 5$, $PQ = 6$
(iii)	$\triangle LMP$: $LM = 2.7$, $MP = 2$, $LP = 3$	$\triangle DEF$: $DE = 4$, $EF = 5$, $DF = 6$
(iv)	$\triangle MNL$: $MN = 2.5$, $ML = 5$, $\angle M = 70°$	$\triangle PQR$: $PQ = 5$, $QR = 10$, $\angle Q = 70°$
(v)	$\triangle ABC$: $AB = 2.5$, $BC = 3$, $\angle A = 80°$	$\triangle DEF$: $DF = 5$, $EF = 6$, $\angle F = 80°$
(vi)	$\triangle DEF$: $\angle D = 70°$, $\angle E = 80°$	$\triangle PQR$: $\angle Q = 80°$, $\angle R = 30°$

SOLUTION

(i) All three pairs of angles are equal. Similar by AAA: $\triangle ABC \sim \triangle PQR$.

(ii) $\frac{AB}{QR} = \frac{2}{4}$, $\frac{BC}{RP} = \frac{2.5}{5}$, $\frac{CA}{PQ} = \frac{3}{6}$. All equal $\frac{1}{2}$. Similar by SSS: $\triangle ABC \sim \triangle QRP$.

(iii) $\frac{MP}{DE} = \frac{2}{4} = 0.5$ and $\frac{LP}{DF} = \frac{3}{6} = 0.5$, but $\frac{LM}{EF} = \frac{2.7}{5} = 0.54$. The sides are not proportional, so the triangles are not similar.

(iv) $\frac{MN}{QP} = \frac{2.5}{5} = \frac{1}{2}$ and $\frac{ML}{QR} = \frac{5}{10} = \frac{1}{2}$, and the included angles $\angle M$ and $\angle Q$ are both $70°$. Similar by SAS: $\triangle MNL \sim \triangle QPR$.

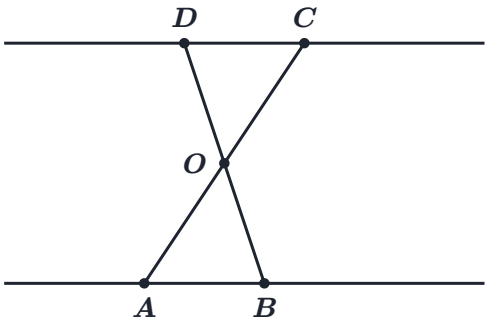
(v) $\frac{AB}{DF} = \frac{BC}{EF} = \frac{1}{2}$, but $\angle A$ is not the angle between AB and BC (that is $\angle B$), so SAS cannot be used. Not similar.

(vi) $\angle F = 180° - 70° - 80° = 30°$ and $\angle P = 180° - 80° - 30° = 70°$. The angles match. Similar by AAA: $\triangle DEF \sim \triangle PQR$.

Answer: similar in (i), (ii), (iv) and (vi). Not similar in (iii) and (v).

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Q2 In the figure, $\triangle ODC \sim \triangle OBA$, $\angle BOC = 125°$ and $\angle CDO = 70°$. Find $\angle DOC$, $\angle DCO$ and $\angle OAB$.



SOLUTION

DOB is a straight line, so $\angle DOC = 180° - 125° = 55°$.

In $\triangle DOC$: $\angle DCO = 180° - 70° - 55° = 55°$.

Since $\triangle ODC \sim \triangle OBA$, corresponding angles are equal: $\angle OAB = \angle OCD = 55°$.

Answer: $\angle DOC = 55°$, $\angle DCO = 55°$, $\angle OAB = 55°$

Q3 The diagonals AC and BD of a trapezium $ABCD$ with $AB \parallel DC$ meet at O. Using a similarity criterion, show that $\frac{OA}{OC} = \frac{OB}{OD}$.

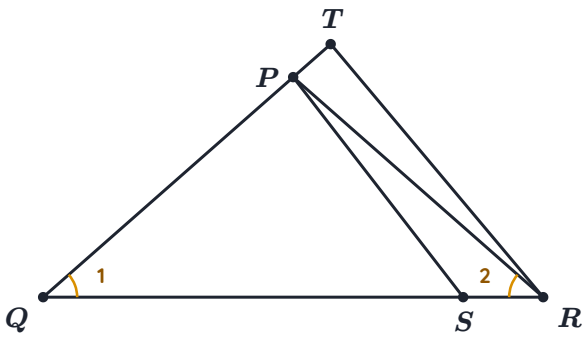
SOLUTION

In $\triangle OAB$ and $\triangle OCD$: $\angle OAB = \angle OCD$ and $\angle OBA = \angle ODC$ (alternate angles, $AB \parallel DC$).

So $\triangle OAB \sim \triangle OCD$ (AA).

Corresponding sides are proportional: $\frac{OA}{OC} = \frac{OB}{OD}$.

Q4 In the figure, S is on QR and T is on QP produced, with $\frac{QR}{QS} = \frac{QT}{PR}$ and $\angle 1 = \angle 2$, where $\angle 1 = \angle PQR$ and $\angle 2 = \angle PRQ$. Show that $\triangle PQS \sim \triangle TQR$.



SOLUTION

In $\triangle PQR$, $\angle 1 = \angle 2$, so the sides opposite them are equal: $PR = PQ$.

The given relation becomes $\frac{QR}{QS} = \frac{QT}{PQ}$, that is $\frac{QS}{QR} = \frac{PQ}{QT}$.

In $\triangle PQS$ and $\triangle TQR$: $\frac{PQ}{TQ} = \frac{QS}{QR}$ and the included angle $\angle Q$ is common.

Hence $\triangle PQS \sim \triangle TQR$ (SAS).

Q5 S and T are points on the sides PR and QR of $\triangle PQR$ such that $\angle P = \angle RTS$. Show that $\triangle RPQ \sim \triangle RTS$.

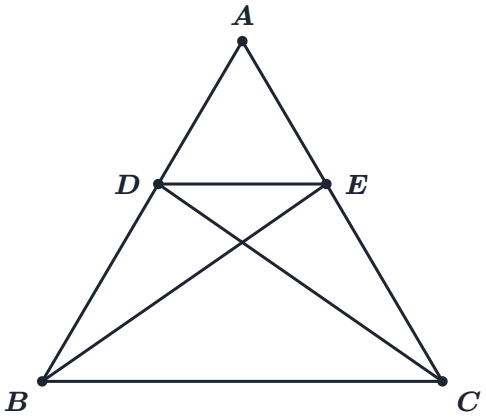
SOLUTION

In $\triangle RPQ$ and $\triangle RTS$: $\angle RPQ = \angle RTS$ (given) and $\angle R$ is common.

Hence $\triangle RPQ \sim \triangle RTS$ (AA).

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Q6 In the figure, D is on AB and E is on AC, and $\triangle ABE \cong \triangle ACD$. Show that $\triangle ADE \sim \triangle ABC$.



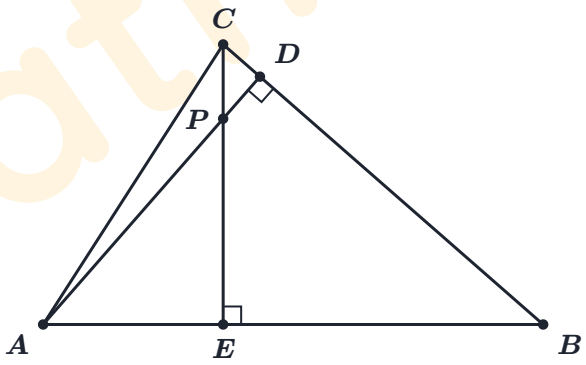
SOLUTION

Since $\triangle ABE \cong \triangle ACD$, the corresponding parts are equal: $AB = AC$ and $AE = AD$.

So $\frac{AD}{AB} = \frac{AE}{AC}$, and $\angle A$ is common to $\triangle ADE$ and $\triangle ABC$.

Hence $\triangle ADE \sim \triangle ABC$ (SAS).

Q7 In the figure, the altitudes AD and CE of $\triangle ABC$ meet at P. Show that (i) $\triangle AEP \sim \triangle CDP$ (ii) $\triangle ABD \sim \triangle CBE$ (iii) $\triangle AEP \sim \triangle ADB$ (iv) $\triangle PDC \sim \triangle BEC$.



SOLUTION

Each pair has a right angle, so one more pair of equal angles gives AA similarity.

(i) $\angle AEP = \angle CDP = 90^\circ$ and $\angle APE = \angle CPD$ (vertically opposite). So $\triangle AEP \sim \triangle CDP$.

(ii) $\angle ADB = \angle CEB = 90^\circ$ and $\angle B$ is common. So $\triangle ABD \sim \triangle CBE$.

(iii) $\angle AEP = \angle ADB = 90^\circ$ and $\angle PAE = \angle BAD$ (the same angle). So $\triangle AEP \sim \triangle ADB$.

(iv) $\angle PDC = \angle BEC = 90^\circ$ and $\angle PCD = \angle BCE$ (the same angle). So $\triangle PDC \sim \triangle BEC$.

All four pairs are similar by AA.

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Q8 E is a point on the side AD produced of a parallelogram $ABCD$, and BE meets CD at F. Show that $\triangle ABE \sim \triangle CFB$.

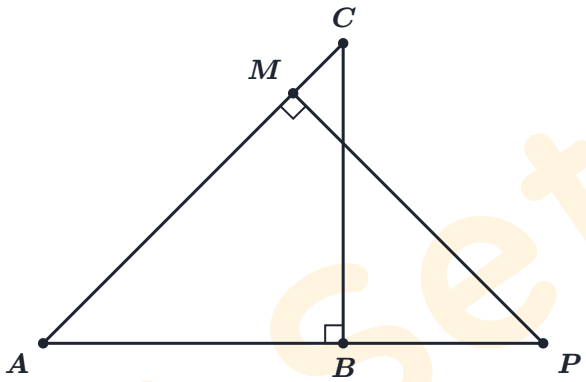
SOLUTION

In $\triangle ABE$ and $\triangle CFB$: $\angle A = \angle C$ (opposite angles of a parallelogram).

$AE \parallel BC$, so $\angle AEB = \angle CBF$ (alternate angles).

Hence $\triangle ABE \sim \triangle CFB$ (AA).

Q9 In the figure, ABC and AMP are right triangles, right-angled at B and M, with M on AC and P on AB produced. Prove that (i) $\triangle ABC \sim \triangle AMP$ (ii) $\dfrac{CA}{PA} = \dfrac{BC}{MP}$.



SOLUTION

(i) In $\triangle ABC$ and $\triangle AMP$: $\angle ABC = \angle AMP = 90^\circ$ and $\angle A$ is common. So $\triangle ABC \sim \triangle AMP$ (AA).

(ii) Corresponding sides of similar triangles are proportional: $\dfrac{CA}{PA} = \dfrac{BC}{MP}$.

Both parts are proved.

Q10 CD and GH are the bisectors of $\angle ACB$ and $\angle EGF$, with D on AB and H on FE, in $\triangle ABC$ and $\triangle EFG$. If $\triangle ABC \sim \triangle FEG$, show that (i) $\dfrac{CD}{GH} = \dfrac{AC}{FG}$ (ii) $\triangle DCB \sim \triangle HGE$ (iii) $\triangle DCA \sim \triangle HGF$.

SOLUTION

$\triangle ABC \sim \triangle FEG$ gives $\angle A = \angle F$, $\angle B = \angle E$ and $\angle ACB = \angle FGE$. Halving the last pair: $\angle ACD = \angle FGH$ and $\angle DCB = \angle HGE$.

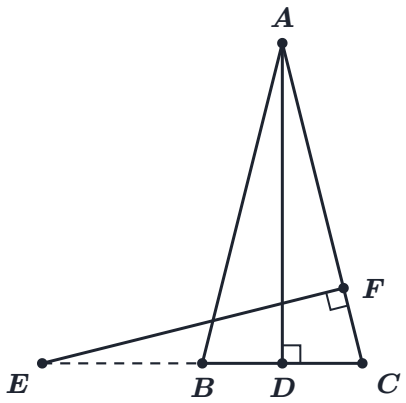
(iii) In $\triangle DCA$ and $\triangle HGF$: $\angle A = \angle F$ and $\angle ACD = \angle FGH$. So $\triangle DCA \sim \triangle HGF$ (AA).

(i) From (iii), $\dfrac{CD}{GH} = \dfrac{AC}{FG}$.

(ii) In $\triangle DCB$ and $\triangle HGE$: $\angle B = \angle E$ and $\angle DCB = \angle HGE$. So $\triangle DCB \sim \triangle HGE$ (AA).

All three parts are proved.

Q11 In the figure, ABC is an isosceles triangle with $AB = AC$, and E is a point on CB produced. If $AD \perp BC$ and $EF \perp AC$, prove that $\triangle ABD \sim \triangle ECF$.



SOLUTION

$AB = AC$, so $\angle ABD = \angle ACB$, that is $\angle ABD = \angle ECF$.

Also $\angle ADB = \angle EFC = 90^\circ$.

Hence $\triangle ABD \sim \triangle ECF$ (AA).

Q12 The sides AB and BC and the median AD of $\triangle ABC$ are proportional to the sides PQ and QR and the median PM of $\triangle PQR$. Show that $\triangle ABC \sim \triangle PQR$.

SOLUTION

Given $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$. Since D and M are mid-points, $\frac{BC}{QR} = \frac{2BD}{2QM} = \frac{BD}{QM}$.

So $\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM}$, and $\triangle ABD \sim \triangle PQM$ (SSS). Hence $\angle B = \angle Q$.

In $\triangle ABC$ and $\triangle PQR$: $\frac{AB}{PQ} = \frac{BC}{QR}$ and the included angles $\angle B$ and $\angle Q$ are equal.

Hence $\triangle ABC \sim \triangle PQR$ (SAS).

Q13 D is a point on the side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB \cdot CD$.

SOLUTION

In $\triangle ADC$ and $\triangle BAC$: $\angle ADC = \angle BAC$ (given) and $\angle C$ is common. So $\triangle ADC \sim \triangle BAC$ (AA).

Corresponding sides are proportional: $\frac{CA}{CB} = \frac{CD}{CA}$.

Hence $CA^2 = CB \cdot CD$.

Q14 The sides AB and AC and the median AD of $\triangle ABC$ are proportional to the sides PQ and PR and the median PM of $\triangle PQR$. Show that $\triangle ABC \sim \triangle PQR$.

SOLUTION

Produce AD to E so that $DE = AD$, and produce PM to N so that $MN = PM$. Join CE and RN.

In $\triangle ABD$ and $\triangle ECD$: $AD = ED$, $BD = CD$ and $\angle ADB = \angle EDC$. So they are congruent (SAS) and $CE = AB$. In the same way $RN = PQ$.

Given $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}$. So $\frac{CE}{RN} = \frac{AC}{PR} = \frac{2AD}{2PM} = \frac{AE}{PN}$, and $\triangle ACE \sim \triangle PRN$ (SSS). Hence $\angle CAD = \angle RPM$.

Joining BE and QN and repeating the argument gives $\angle BAD = \angle QPM$. Adding, $\angle BAC = \angle QPR$.

With $\frac{AB}{PQ} = \frac{AC}{PR}$ and $\angle A = \angle P$, $\triangle ABC \sim \triangle PQR$ (SAS).

Q15 A 6 m tall pole throws a 4 m long shadow. At that moment a tower throws a shadow 28 m long. How tall is the tower?

SOLUTION

At the same time the sun's rays make the same angle with the ground, so the right triangles formed by the pole and the tower with their shadows are similar (AA).

$\frac{\text{height of tower}}{28} = \frac{6}{4}$, so the height is $\frac{6 \times 28}{4} = 42$ m.

Answer: 42 m

Q16 AD and PM are medians of $\triangle ABC$ and $\triangle PQR$, where $\triangle ABC \sim \triangle PQR$. Prove that $\frac{AB}{PQ} = \frac{AD}{PM}$.

SOLUTION

$\triangle ABC \sim \triangle PQR$ gives $\angle B = \angle Q$ and $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{2BD}{2QM} = \frac{BD}{QM}$.

In $\triangle ABD$ and $\triangle PQM$: $\frac{AB}{PQ} = \frac{BD}{QM}$ and $\angle B = \angle Q$. So $\triangle ABD \sim \triangle PQM$ (SAS).

Hence $\frac{AB}{PQ} = \frac{AD}{PM}$.

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